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United States Patent	12399075
Kind Code	B2
Date of Patent	August 26, 2025
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Pipeline pressure monitoring device

Abstract

A pipeline pressure monitoring device includes a housing, a vent nozzle, a first circuit board, a pressure sensor, a sliding sleeve, an elastic member sleeved on the vent nozzle and sandwiched between the first stop block and the sliding sleeve, a magnetic block on the sliding sleeve, a second circuit board connected to the first circuit board, and a position sensor on the second circuit board and cooperating with the magnetic block. A first stop block is arranged on the vent nozzle. One end of the vent nozzle protrudes from the housing through a mounting hole of the housing. The pressure sensor is installed at the other end of the vent nozzle and on the first circuit board. The pressure sensor blocks the other end of the vent nozzle. The sliding sleeve is installed in the mounting hole in a sliding manner and sleeved on the vent nozzle.

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Appl. No.:	18/037485
Filed (or PCT Filed):	December 31, 2020
PCT No.:	PCT/CN2020/141900
PCT Pub. No.:	WO2022/105036
PCT Pub. Date:	May 27, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240011860 A1	Jan. 11, 2024

Foreign Application Priority Data

Publication Classification

Int. Cl.: A61M5/168 (20060101); G01L19/00 (20060101); G01L19/14 (20060101)

U.S. Cl.:

CPC G01L19/0038 (20130101); A61M5/16854 (20130101); G01L19/142 (20130101);

Field of Classification Search

CPC: G01L (19/0038); G01L (19/142); G01L (11/00); G01L (19/143); G01L (19/003); A61M (5/16854); G01B (7/003)

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to a medical device, and more particularly to a pipeline pressure monitoring device.

BACKGROUND OF THE INVENTION

(2) In medical device applications, especially blood processing equipment and blood component collection equipment, it is necessary to monitor the pressure of a system. While monitoring the pressure, it is necessary to monitor the installation of consumables. The mechanical structure of a conventional pressure monitoring device includes a guide cap, a guide cap fixing seat, a guide cap bracket, an elastic ring, an elastic ring fixing nut, a conical vent nozzle, a sealing ring, and a bracket. The guide cap is made of a plastic material. The elastic ring is made of rubber. The elastic ring is fitted on the outside of the conical vent nozzle and has a concave groove. The elastic ring is tightened and pressed on the conical vent nozzle by the elastic ring fixing nut. The guide cap is inserted into the concave groove of the elastic ring and fixed on the guide cap fixing seat. Due to its complex mechanical structure and materials, it has the following characteristics: complex structure, large volume, poor smoothness and short service life and it is damaged easily.

SUMMARY OF THE INVENTION

(3) The present invention is to solve the above problems. The primary object of the present invention is to provide a pipeline pressure monitoring device, which has simple structure, small size, good smoothness and long service life and is not easily damaged

(4) The present invention provides a pipeline pressure monitoring device, comprising: a housing, one side of the housing having a mounting hole; a vent nozzle, the vent nozzle having a vent, the vent nozzle being installed in the housing, one end of the vent nozzle extending out of the housing via the mounting hole, a first stop block being provided on the vent nozzle; a first circuit board, installed in the housing; a pressure sensor, configured to detect pressure inside the vent nozzle, the pressure sensor being installed on the first circuit board and connected to the first circuit board, the pressure sensor being matched with another end of the vent nozzle facing an inside of the housing, the pressure sensor being installed at the another end of the vent nozzle facing the inside of the housing to block the another end; a sliding sleeve, installed in the mounting hole in a sliding manner, the sliding sleeve having a through hole, the sliding sleeve being sleeved on the vent nozzle via the through hole, a limit member being provided on the sliding sleeve, the limit member being located inside the housing; an elastic member, sleeved on the vent nozzle and sandwiched between the first stop block and the sliding sleeve; a magnetic block, installed on the sliding sleeve; a second circuit board, installed in the housing and connected to the first circuit board; and a position sensor, installed on the second circuit board and connected to the second circuit board, wherein the installation position of the position sensor meets that: after a consumable connector is installed on the vent nozzle, the distance that the sliding sleeve slides toward the inside of the housing is just enough to make the magnetic block move into a detection area of the position sensor.

(5) Furthermore, in the pipeline pressure monitoring device provided by the present invention, a sealing ring is provided between the pressure sensor and the vent nozzle.

(6) Furthermore, in the pipeline pressure monitoring device provided by the present invention, the elastic member is a spring.

(7) Furthermore, in the pipeline pressure monitoring device provided by the present invention, a bushing is provided between the sliding sleeve and the vent nozzle, the bushing is fixed on the through hole of the sliding sleeve, and the bushing is slidably sleeved on the vent nozzle.

(8) Furthermore, in the pipeline pressure monitoring device provided by the present invention, the cross-section of the end of the vent nozzle facing an outside of the housing gradually decreases toward the outside of the housing, the cross-section of the vent is constant, a section of the vent nozzle that is gradually tapered does not contact the through hole, and the bushing is located on one side of the through hole facing the inside of the housing.

(9) The present invention provides the following advantages:

(10) The present invention has simple structure, small size, good smoothness and long service life, and is not easily damaged

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of the pipeline pressure monitoring device according to an embodiment of the present invention; and

(2) FIG. 2 is a cross-sectional view of the pipeline pressure monitoring device according to the embodiment of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(3) For the technical means, creative features, objects and effects achieved by the present invention to be understood easily, the following embodiments will specifically illustrate the pipeline pressure monitoring device of the present invention, with reference to the accompanying drawings.

(4) As shown in FIG. 1 and FIG. 2, the pipeline pressure monitoring device **100** comprises a housing **10**, a vent nozzle **20**, a first circuit board **30**, a pressure sensor **40**, a sliding sleeve **50**, an elastic member **60**, a magnetic block **70**, a second circuit board **80**, and a position sensor **90**.

(5) One side of the housing **10** has a mounting hole **11**.

(6) The vent nozzle **20** has a vent **21**. The vent nozzle **20** is installed in the housing **10**. One end of the vent nozzle **20** extends out of the housing **10** via the mounting hole **11**. A first stop block **22** is provided on the vent nozzle **20**.

(7) The first circuit board **30** is installed in the housing **10**. The pressure sensor **40** is configured to detect the pressure inside the vent nozzle **20**. The pressure sensor **40** is installed on the first circuit board **30** and connected to the first circuit board **30**. The pressure sensor **40** is matched with one end of the vent **21** facing the inside of the housing **10**. The pressure sensor **40** is installed at the end of the vent **21** facing the inside of the housing **10** to block the end. After the vent nozzle **20** is installed on the pressure sensor **40**, the first circuit board **30** and the pressure sensor **40** play a role in fixing the vent nozzle **20**.

(8) In this embodiment, a sealing ring **41** is provided between the pressure sensor **40** and the vent nozzle **20**.

(9) The sliding sleeve **50** is installed in the mounting hole **11** in a sliding manner. The sliding sleeve **50** has a through hole **51**. The sliding sleeve **50** is sleeved on the vent nozzle **20** via the through hole **51**. A limit member **52** is provided on the sliding sleeve **50**. The limit member **52** is located inside the housing **10**, and is configured to prevent the sliding sleeve **50** from sliding out of the mounting hole **11**.

(10) In the present embodiment, a bushing **53** is provided between the sliding sleeve **50** and the vent nozzle **20**. The bushing **53** is fixed on the through hole **51** of the sliding sleeve **50**. The bushing **53** is slidably sleeved on the vent nozzle **20**.

(11) In this embodiment, the cross-section of the other end of the vent nozzle **20** facing the outside of the housing **10** gradually decreases toward the outside of the housing **10**, and the cross-section of the vent **21** is constant, that is, referring to FIG. 2, the wall at the left end of the vent nozzle **20** gradually becomes thinner towards the left. Specifically, the end of the vent nozzle **20** facing the outside of the housing **10** is a hollow cylinder. The outer diameter of the hollow cylinder gradually decreases toward the outside of the housing **10**, while the inner diameter is constant. A section of the vent nozzle **20** that is gradually tapered does not contact the through hole **51**. The bushing **52** is located on one side of the through hole **51** facing the inside of the housing **10**. The elastic member **60** is sleeved on the vent nozzle **20** and sandwiched between the first stop block **22** and the sliding sleeve **50**. When no consumable connector is installed on the vent **20**, the elastic member **60** is in a free or slightly compressed state. When the elastic member **60** is in a free state, both ends of the elastic member **60** are in contact with the first stop block **22** and the sliding sleeve **50**, respectively.

(12) In this embodiment, the elastic member **60** is a spring.

(13) The magnetic block **70** is installed on the sliding sleeve **50**. Specifically, the magnetic block **70**

is mounted on the sliding sleeve **50** inside the housing **10**.

(14) The second circuit board **80** is installed in the housing **10**, and is connected to the first circuit board **30**.

(15) The position sensor **90** is installed on the second circuit board **80**, and is connected to the second circuit board **80**. The installation position of the position sensor **90** meets that: after the consumable connector is installed on the vent nozzle **20**, the distance that the sliding sleeve **50** slides toward the inside of the housing **10** is just enough to make the magnetic block **70** move into the detection area of the position sensor **90**. Specifically, the position sensor **90** is Hall sensor. When the magnetic block **70** moves into the detection area of the position sensor **90**, the position sensor **90** outputs 1; when the magnetic block **70** is not in the detection area of the position sensor **90**, the position sensor **90** outputs 0.

(16) Both the first circuit board **30** and the second circuit board **80** are connected with the external equipment, and are configured to send the detected pressure information and the signal detected by the position sensor **90** to the external equipment, and the detection data is displayed by the external equipment.

(17) When the consumable connector is installed on the vent nozzle **30**, the consumable connector will push the sliding sleeve **50** to move toward the inside of the housing **10** and compress the elastic member **60**. The magnetic block **70** moves toward the inside of the housing **10** along with the sliding sleeve **50**. After the consumable connector is installed on the vent nozzle **30**, the magnetic block **70** moves to the detection area of the position sensor **90**. After the magnetic block **70** is sensed by the position sensor **90**, the output signal changes. Therefore, it can be judged whether the consumable connector is installed in the vent nozzle **30** correctly. The pressure sensor **40** detects the pressure of the gas in the vent nozzle **30**, so that the pressure of the gas in the pipeline of the consumable connector connected to the vent nozzle **30** and the equipment connected to the consumable connector can be obtained. After the consumable connector is taken out, under the elastic force of the elastic member **60**, the sliding sleeve **50** is returned.

(18) The above embodiment is a preferred embodiment of the present invention, and is not intended to limit the protection scope of the present invention.

Claims

1. A pipeline pressure monitoring device, comprising: a housing, one side of the housing having a mounting hole; a vent nozzle, the vent nozzle having a vent, the vent nozzle being installed in the housing, one end of the vent nozzle extending out of the housing via the mounting hole, a first stop block being provided on the vent nozzle; a first circuit board, installed in the housing; a pressure sensor, configured to detect pressure inside the vent nozzle, the pressure sensor being installed on the first circuit board and connected to the first circuit board, the pressure sensor being matched with another end of the vent nozzle facing an inside of the housing, the pressure sensor being installed at the another end of the vent nozzle facing the inside of the housing to block the another end; a sliding sleeve, installed in the mounting hole in a sliding manner, the sliding sleeve having a through hole, the sliding sleeve being sleeved on the vent nozzle via the through hole, a limit member being provided on the sliding sleeve, the limit member being located inside the housing; an elastic member, sleeved on the vent nozzle and sandwiched between the first stop block and the sliding sleeve; a magnetic block, installed on the sliding sleeve; a second circuit board, installed in the housing and connected to the first circuit board; and a position sensor, installed on the second circuit board and connected to the second circuit board, wherein the installation position of the position sensor meets that: after a consumable connector is installed on the vent nozzle, the distance that the sliding sleeve slides toward the inside of the housing is just enough to make the magnetic block move into a detection area of the position sensor.

2. The pipeline pressure monitoring device as claimed in claim 1, wherein a sealing ring is

provided between the pressure sensor and the vent nozzle.

3. The pipeline pressure monitoring device as claimed in claim 1, wherein the elastic member is a spring.

4. The pipeline pressure monitoring device as claimed in claim 1, wherein a bushing is provided between the sliding sleeve and the vent nozzle, the bushing is fixed on the through hole of the sliding sleeve, and the bushing is slidably sleeved on the vent nozzle.

5. The pipeline pressure monitoring device as claimed in claim 4, wherein the cross-section of the end of the vent nozzle facing an outside of the housing gradually decreases toward the outside of the housing, the cross-section of the vent is constant, a section of the vent nozzle that is gradually tapered does not contact the through hole, and the bushing is located on one side of the through hole facing the inside of the housing.
